SPNE Check

First, we write an expression for cumulative reward for the agent assuming both follow their strategy fully

Let, V_P be the cumulative reward for the agent following the strategy. u or more precisely, $u_1(a_1, a_2)$ be the payoff the agent 1 receives, when the action of each agent is a_1 and a_2 respectively. p is the probability of collapse into degraded state. p is also a function of actions of both agents.

m is the penalty in the degraded state δ be the discount factor.

To derive expression for V_P, we can write the following equation,

$$V_P = (1-p) * (\text{cumulative reward in prosperous state}) + p * (\text{cumulative reward in degraded state}) \quad (1)$$

Now, immediate reward in prosperous state is u + discounted cumulative future reward in prosperous state, which is δV_P. Likewise, immediate reward in degraded state is m + discounted cumulative future reward in degraded state, which is δV_D.

So, we can write,

$$V_P = (1-p)(u + \delta V_P) + p(m + \delta V_D) \quad (2)$$

V_D is simpler to derive.

$$V_D = m + \delta V_D \implies V_D = \frac{m}{1-\delta} \quad (3)$$

Substituting the value of V_D in the equation for V_P, we get,

$$\begin{aligned} V_P &= (1-p)(u + \delta V_P) + p\left(m + \delta \frac{m}{1-\delta}\right) \\ \implies V_P &= (1-p)(u + \delta V_P) + \frac{pm}{1-\delta} \\ \implies V_P - (1-p)\delta V_P &= (1-p)u + \frac{pm}{1-\delta} \\ \implies V_P(1 - (1-p)\delta) &= (1-p)u + \frac{pm}{1-\delta} \\ \implies V_P &= \frac{(1-p)u + \frac{pm}{1-\delta}}{1-(1-p)\delta} \end{aligned} \quad (4)$$

This can also be written as,

$$\boxed{V_P = \frac{u}{1-\delta} + \frac{p(m-u)}{(1-\delta)[1-\delta(1-p)]}} \quad (5)$$

Now, let us consider what happens if agent 1 deviates from the strategy, at time $t = 0$, WLG. The payoff then becomes u', which is the payoff for the action profile where agent 1 deviates and agent 2 follows the strategy. The probability of collapse for that time step becomes p', because in the EcoPG risk of collapse is dependent on the joint action at each time step.

The strategy is SPNE if the payoff from deviation V'_P is less than or equal to the payoff from following the strategy V_P.

$$V_P = (1-p')(u' + \delta V_P(t=1)) + p'(m + \delta V_D) \quad (6)$$

p' and u' above can also be written as $p(t=0)$ and $u(t=0)$ respectively

This represents the cumulative reward for agent 1 due to single step deviation. Note that the logic follows 2, with approriate substitutions for the payoff and probability of collapse due to the single step deviation.

However, note:

a. $V_P(t=1)$ is the cumulative reward for agent 1 from time step $t=1$ onwards. Now, if there are no future consequences of the deviation, in terms of what actions agents take in the future (e.g., when they both are playing memory-0 strategies), then $V_P(t=1)$ will be the same as V_P.

b. Consider a strategy like WSLS and assume both cooperate from the beginning. If agent 1 deviates at $t=0$ ((D,C) at $t=0$), agent 1 and agent 2 will defect at $t=1$ ((D,D) at $t=1$). By $t=2$, both will again cooperate ((C,C) for all $t \geq 2$).

c. consider a strategy like GT and assume both cooperate from the beginning. If agent 1 deviates at $t=0$ ((D,C) at $t=0$), then agent 2 will punish at $t=1$ by defecting (joint action (D,D) at t $=1$), and both will consequently keep defecting forever ($(D,D)\forall t \geq 2$ ..).

d. In both cases (b) and (c), although the deviation is single step, there are future consequences of the deviation in terms of actions taken by both agents in the future. This also affects the probability of collapse as well in the subsequent time steps. In case, (b), the effect is temporary, and at $t=2$, things return to normalcy. But for case (c), the effect is permanent.

$$V_P(t=1) = (1-p(t=1))(u(t=1)+\delta V_P(t=2)) + p(t=1)(m+\delta V_D) \tag{7}$$

$$V_P(t=2) = (1-p(t=2))(u(t=2)+\delta V_P(t=2)) + p(t=2)(m+\delta V_D) \tag{8}$$

For memory-0 strategies, $u(t=1)=u(t=2)=u$. Likewise, $p(t=1)=p(t=2)$ and $V_P(t=1)=V_P(t=2)$. In this case, we do not need to compute $V_P(t=1)$ and $V_P(t=2)$ separately, since both are equal to V_P.

For WSLS-like strategies, $u(t=1) \neq u(t=0)$, but $u(t=2)=u$. Likewise, $p(t=1)$ may differ, but $p(t=2)=p$, and similarly $V_P(t=2)=V_P$. Thus, we need to compute $V_P(t=1)$ separately, while $V_P(t=2)$ is equal yp V_P.

For GT-like strategies, $u(t=2) \neq u(t=1)=u(t=0)$. Likewise, $p(t=1)$ and $p(t=2)$ differ, and $V_P(t=1)$ and $V_P(t=2)$ are also distinct. Therefore, both $V_P(t=1)$ and $V_P(t=2)$ must be computed separately, and neither equals V_P.

As we don't go beyond memory-1 strategies, there is no need to continue this recursion to calculate V_P beyond $t=2$. We only need to calculate V_P at $t=0$, $t=1$, and $t=2$, since $t=0$ is the deviation step, $t=1$ is the reaction (if any) which can change V_P, and $t=2$ is the change because of the reaction due to deviation (if any).

After, $t=2$, the action sequence is along their strategy profile given what happened at t $=2$, so $V_P(t=3)=V_P(t=2)$.

Now, consider the special case of a strategy lik Tit-for-tat (TFT). Here, while both play the strategy, there is continuous mutual cooperation. However, if there is unilateral defection: at $t=0$, agent 1 defects C,D, then according to TFT, at $t=1$, the joint action is D,C. Then, it's C,D at $t=2$ again leading to a endless alternating between C,D and D,C.

For this case also, we can write an expression for cumulative rewards upon deviation: but we will revisit this again if needed.